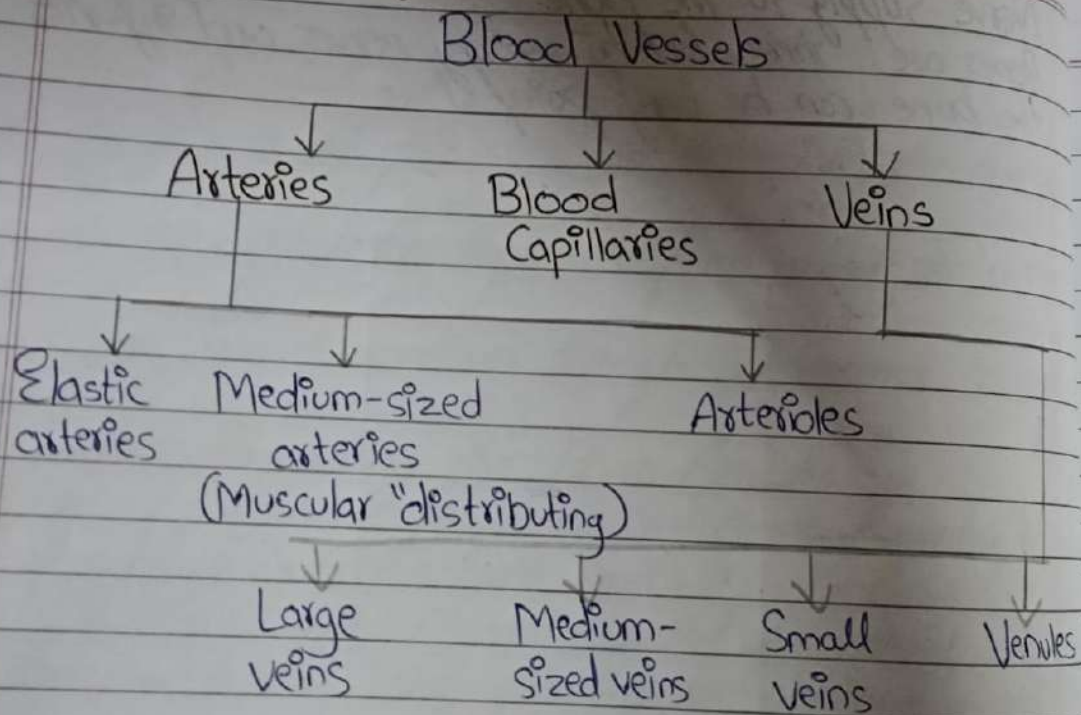


Histology of Blood Vessels.

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The wall of concentric

Tunica interna (interna)

It is the innermost layer of:

Muscular arteries (Medium-sized artery)

Example: brachial, ulnar, renal, Femoral.

Tunica intima (interna)	Tunica media	Tunica adventitia (externa)
• Endothelium • Subendothelial C.T. layer • Internal elastic lamina: Is prominent and displays an undulating surface	(Thicker than T.A. or similar in thickness) Components: A. Smooth muscle cell (SMCs) "predominant component" B. In between the Smooth muscle, there are: 1- Elastic fibers 2- Collagen fibers (type I) 3- Reticular fibers (type III collagen) C. External elastic lamina: may be identifiable.	• loose C.T.

Veins (Medium-sized vein)

Thickness of the wall: thinner than the accompanying artery. wide lumen, thick wall.

Tunica intima (interna)	Tunica media	Tunica adventitia (externa)
• Usually forms valves • No elastic internal lamina • endothelium	Consists of: 1. Fewer SMCs. 2. Types I & III collagen fibers	• thicker than T. media

Veins

Medium-Sized Vein.

* Valves of Veins

- Valve of a vein is composed of 2 leaflets
- Each leaflet has a thin fold of the T. intima
- Components:
 - Endothelium
 - Core of C.T.

Blood Capillaries.

Diameter: usually 8-10 μm .

Microscopic structure:

1. Single layer of squamous endothelial cells.
2. Basal lamina: Surrounds the external surface of the endothelial cells.
3. Pericytes: Have processes, share the basal lamina of the endothelial cells.

Types:

1. Continuous blood capillaries.
2. Fenestrated blood capillaries.
 - a. With diaphragms
 - b. Without diaphragms.
3. Sinusoidal blood capillaries.

Arteries.

Elastic arteries (Large artery)

Example: Aorta, common carotid artery, subclavian artery, Common iliac artery, Pulmonary trunk.

Tunica intima (interna)	Tunica media	Tunica adventitia (externa)
• Endothelium • Subendothelial C.T. • Internal elastic lamina: not prominent "not clear" & indistinct.	It consists of: A. Fenestrated elastic membranes: sheets & lamellae "main component of T.M." B. In between there are: 1- Elastic fibers "predominant (main) component = 90%" 2- Collagen fibers (type I collagen) 3- Reticular fibers (type III collagen) 4- Smooth muscle cells Much thicker than T.I and T.A.	• Much thinner than T.M. • It is composed of loose C.T. • Contains Vasa Vasorum → Send branches to the outer part of T.M.



The wall of the blood vessel is formed of three concentric layers:-

Tunica intima (interna)	Tunica media	Tunica adventitia (externa)
It is the innermost layer composed of:	Intermediate layer Composed of:	Outermost layer Composed of connective tissue containing Vasa Vasorum:
* Endothelial cells: Simple Squamous epithelium. * Subendothelial layer: loose C.T. * Internal elastic lamina: fenestrated elastic sheet. For the diffusion of nutrients from the blood to the vessel.	1. Mainly Smooth muscles arranged helically 2. Elastic fibres 3. Type III collagen (reticular fibers). 4. Type I collagen. NB: large muscular arteries have external elastic lamina, separating the tunica media from the tunica adventitia.	* They are small arterioles in tunica adventitia and the outer part of tunica media. * They are more prevalent in the walls of veins than arteries why? Venous blood contains less oxygen and nutrients than arterial blood.